TabuLLM: AI Benchmark Report

1. Executive Summary: Consolidated Audit Results

This report consolidates the results of the TabuLLM data augmentation pipeline. The objective is to audit the fidelity, diversity, and privacy of the synthetic data generated via LLM inference. Below are the averaged performance indicators across all processed datasets in this session.

Audit Indicator	Global Session Mean (± Std)
Synthetic Fidelity (JSD)	79.0 ± 2.61
Correlation Stability	87.7 ± 2.37
Distribution Consistency	86.5 ± 2.42
Entropy Preservation	88.5 ± 0.81
Mean Z Score Drift	0.2 ± 0.03
Categorical Fidelity	82.8 ± 0.60
Privacy Risk	49.5 ± 1.20
Data Density	99.9 ± 0.21
Coverage Score	19.3 ± 5.18
Feature Accuracy	78.6 ± 2.29
Novelty Score	0.4 ± 0.01

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Dataset Audit: indian_liver_patient_dataset.csv

1. Dataset Portrait

Attribute	Value
Rows	583
Columns	11
Numeric features	10
Categorical features	1
Missing cells (%)	0.1%

2. Generation Strategy & LLM Config [1]

Parameter	Configuration
Algorithm (*12)	gpt-4o-mini
Folds (*13)	10
Context % (*14)	20%
Generation % (*15)	100%
Join context (*16)	True

3. Statistical Significance & Data Drift

Feature	Orig Mean	Aug Mean	Shift	P-Value	Status
Age	44.746	45.273	+0.526	1.2%	OK
TB	3.299	1.718	-1.581	47.9%	Drift
DB	1.486	0.704	-0.782	52.7%	Drift
Alkphos	290.576	217.172	-73.405	25.3%	Drift
Sgpt	80.714	41.388	-39.325	48.7%	Drift
Sgot	109.911	46.925	-62.986	57.3%	Drift
TP	6.483	6.730	+0.246	3.8%	OK
ALB	3.142	3.401	+0.259	8.2%	OK
A/G Ratio	0.947	1.002	+0.055	5.8%	OK
Selector	1.286	1.333	+0.046	3.6%	OK

4. Quality & Fidelity Metrics (Mean ± Std)

Metric Category	Aggregated Score
Synthetic Fidelity (JSD) (*1)	79.0 ± 2.61
Correlation Stability (*2)	87.7 ± 2.37
Distribution Consistency (*3)	86.5 ± 2.42
Entropy Preservation (*4)	88.5 ± 0.81
Mean Z Score Drift (*5)	0.2 ± 0.03
Categorical Fidelity (*6)	82.8 ± 0.60
Privacy Risk (*7)	49.5 ± 1.20
Data Density (*8)	99.9 ± 0.21
Coverage Score (*9)	19.3 ± 5.18

Feature Accuracy (*10)	78.6 ± 2.29
Novelty Score (*11)	0.4 ± 0.01

*[1] (*12) Algorithm: Contextual constraint or model parameter.*

*[2] (*13) Folds: Contextual constraint or model parameter.*

*[3] (*14) Context %: Contextual constraint or model parameter.*

*[4] (*15) Generation %: Contextual constraint or model parameter.*

*[5] (*16) Join context: Contextual constraint or model parameter.*

[6] Synthetic Fidelity (JSD): Overall distribution similarity.
[7] Correlation Stability: Multivariate relationships preserved.
[8] Distribution Consistency: Median feature similarity.
[9] Entropy Preservation: Information diversity maintained.
[10] Mean Z Score Drift: Standardized distribution shift.
[11] Categorical Fidelity: Category distribution similarity.
[12] Privacy Risk: Synthetic rows too close to originals.
[13] Data Density: Non-missing cells in augmented data.
[14] Coverage Score: Synthetic coverage of original space.
[15] Feature Accuracy: Mean feature similarity.
[16] Novelty Score: Distance ratio to nearest neighbors.

6. Downstream ML Utility: Comparative Analysis

Model: XGB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.7025641025641025	0.5880341880341885	0.7008547008547008	-0.00170940170940170 33	
F1	0.7997903571549948	0.6968960212044484	0.7959778877430375	-0.00381246941195770 86	

Model: DT

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.6538461538461545	0.5410256410256411	0.6692307692307693	0.01538461538461537 5	
F1	0.7590596522904013	0.6489367340987894	0.7701827693470424	0.01112311705664098 5	

Model: NB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.5589743589743595	0.5931623931623931	0.5700854700854701	0.01111111111111117	
F1	0.5587432119380917	0.6938541586428602	0.5792288369924201	0.02048562505432841 7	*

Model: MLP

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.7094017094017093	0.5829059829059828	0.6991452991452991	-0.01025641025641025	
F1	0.8133447322663196	0.7048195605170251	0.8120243507406372	-0.00132038152568242 36	

Model: LR

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.7059829059829059	0.5803418803418803	0.7076923076923076	0.00170940170940172 54	
F1	0.8128002697439015	0.6848960144226369	0.8152200372554559	0.00241976751155439 36	

Model: SVM

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Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.7068376068376068	0.6811965811965812	0.7068376068376068	0	
F1	0.8279091624529537	0.790553732351688	0.8279091624529537	0	

Model: RF

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.7034188034188034	0.5880341880341879	0.7017094017094017	-0.00170940170940171	
F1	0.8082421704426773	0.7042492837728536	0.8088558515702239	0.00061368112754666	

Model: KNN

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.6863247863247863	0.6512820512820513	0.6888888888888889	0.00256410256410255	
F1	0.7971299319768493	0.7595425240074631	0.7983264995810337	0.00119656760418443	

Model: AB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.6743589743589744	0.593162393162393	0.6991452991452991	0.02478632478632479	
F1	0.7821343746683586	0.7114298057704651	0.8056272334748323	0.02349285880647366	

Model: GB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.6820512820512821	0.5683760683760684	0.6854700854700855	0.00341880341880341	
F1	0.7886236381078809	0.6809513779937613	0.7865262778875164	-0.00209736022036449	

Appendix: Benchmark Data Trace

Legend: [Blue] = Real Context | [Green] = Generated Synthetic | [Yellow] = Real Training (Remaining) | [Grey] = Real Test Baseline

A. Baseline: Original Dataset (Full Record Sample)

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	65	Female	0.7	0.1	187	16	18	6.8	3.3	0.9
2	62	Male	10.9	5.5	699	64	100	7.5	3.2	0.74
3	62	Male	7.3	4.1	490	60	68	7.0	3.3	0.89
4	58	Male	1.0	0.4	182	14	20	6.8	3.4	1.0
5	72	Male	3.9	2.0	195	27	59	7.3	2.4	0.4
6	46	Male	1.8	0.7	208	19	14	7.6	4.4	1.3
7	26	Female	0.9	0.2	154	16	12	7.0	3.5	1.0
8	29	Female	0.9	0.3	202	14	11	6.7	3.6	1.1
9	17	Male	0.9	0.3	202	22	19	7.4	4.1	1.2
10	55	Male	0.7	0.2	290	53	58	6.8	3.4	1.0
11	57	Male	0.6	0.1	210	51	59	5.9	2.7	0.8
12	72	Male	2.7	1.3	260	31	56	7.4	3.0	0.6
13	64	Male	0.9	0.3	310	61	58	7.0	3.4	0.9
14	74	Female	1.1	0.4	214	22	30	8.1	4.1	1.0
15	61	Male	0.7	0.2	145	53	41	5.8	2.7	0.87
16	25	Male	0.6	0.1	183	91	53	5.5	2.3	0.7
17	38	Male	1.8	0.8	342	168	441	7.6	4.4	1.3
18	33	Male	1.6	0.5	165	15	23	7.3	3.5	0.92
19	40	Female	0.9	0.3	293	232	245	6.8	3.1	0.8
20	40	Female	0.9	0.3	293	232	245	6.8	3.1	0.8

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B.1 Audit Trace: Fold 1 | Ctx: 20% | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	58	Male	1.0	0.5	158	37	43	7.2	3.6	1.0
2	46	Male	1.8	0.7	208	19	14	7.6	4.4	1.3
3	55	Male	3.6	1.6	349	40	70	7.2	2.9	0.6
4	25	Female	0.9	0.3	159	24	25	6.9	4.4	1.7
5	26	Male	1.3	0.4	173	38	62	8.0	4.0	1.0
6	64	Female	0.8	0.2	178	17	18	6.3	3.1	0.9
7	34	Female	0.9	0.3	150	30	40	6.5	3.2	1.0
8	45	Male	1.5	0.6	200	50	60	7.0	4.0	1.2
9	29	Female	0.7	0.2	180	25	30	6.8	3.5	1.1
10	52	Male	2.1	1.0	220	75	80	7.5	4.5	1.3
11	38	Female	1.0	0.4	160	35	45	6.7	3.8	1.2
12	60	Male	3.0	1.5	250	90	100	8.0	4.2	1.5
13	33	Male	7.1	3.7	196	622	497	6.9	3.6	1.09
14	38	Male	2.2	1.0	310	119	42	7.9	4.1	1.0
15	64	Male	3.0	1.4	248	46	40	6.5	3.2	0.9
16	40	Female	0.9	0.2	285	32	27	7.7	3.5	0.8

B.2 Audit Trace: Fold 2 | Ctx: 20% | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	49	Male	1.0	0.3	230	48	58	8.4	4.2	1.0
2	33	Male	0.9	0.8	680	37	40	5.9	2.6	0.8
3	60	Male	3.2	1.8	750	79	145	7.8	3.2	0.69
4	40	Female	2.1	1.0	768	74	141	7.8	4.9	1.6
5	65	Male	4.9	2.7	190	33	71	7.1	2.9	0.7
6	58	Male	0.9	0.2	1100	25	36	7.1	3.5	0.9
7	29	Female	1.2	0.5	210	35	40	6.5	3.0	0.8
8	53	Male	2.5	1.2	300	45	50	7.0	3.5	0.9
9	39	Female	0.8	0.3	180	30	25	6.0	2.8	0.7
10	62	Male	4.0	2.0	400	70	80	7.5	4.0	1.0
11	45	Female	1.5	0.7	250	40	35	6.8	3.2	0.9
12	34	Male	2.1	1.0	320	60	55	7.2	3.8	1.1
13	50	Female	1.0	0.3	191	22	31	7.8	4.0	1.0
14	52	Female	0.6	0.1	194	10	12	6.9	3.3	0.9
15	30	Male	0.8	0.2	182	46	57	7.8	4.3	1.2
16	58	Male	0.8	0.2	298	33	59	6.2	3.1	1.0

B.3 Audit Trace: Fold 3 | Ctx: 20% | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	51	Male	0.8	0.2	230	24	46	6.5	3.1	nan
2	19	Female	0.7	0.2	186	166	397	5.5	3.0	1.2
3	70	Male	1.4	0.6	146	12	24	6.2	3.8	1.58
4	18	Male	1.4	0.6	215	440	850	5.0	1.9	0.6
5	60	Male	2.3	0.6	272	79	51	6.6	3.5	1.1
6	33	Male	0.8	0.2	135	30	29	7.2	4.4	1.5
7	29	Female	1.2	0.4	200	45	50	6.5	3.4	1.1
8	37	Male	0.9	0.3	180	30	25	7.0	3.5	1.0
9	54	Female	2.1	0.8	210	60	70	6.8	3.0	0.9
10	41	Male	1.5	0.5	250	50	60	7.2	3.6	1.2
11	62	Female	0.7	0.2	160	20	30	6.0	3.1	1.0
12	34	Male	3.0	1.0	220	80	90	6.7	3.2	1.1
13	44	Female	1.9	0.6	298	378	602	6.6	3.3	1.0
14	30	Male	0.8	0.2	182	46	57	7.8	4.3	1.2
15	53	Male	0.9	0.4	238	17	14	6.6	2.9	0.8
16	28	Male	0.8	0.3	190	20	14	4.1	2.4	1.4

B.4 Audit Trace: Fold 4 | Ctx: 20% | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	14	Male	1.4	0.5	269	58	45	6.7	3.9	1.4
2	26	Male	2.0	0.9	157	54	68	6.1	2.7	0.8
3	24	Male	0.7	0.2	218	47	26	6.6	3.3	1.0
4	45	Male	0.8	0.2	140	24	20	6.3	3.2	1.0
5	38	Male	2.7	1.4	105	25	21	7.5	4.2	1.2
6	25	Male	0.6	0.1	183	91	53	5.5	2.3	0.7
7	22	Female	0.9	0.3	180	28	30	6.5	3.2	1.1
8	34	Male	1.5	0.6	210	45	50	7.0	3.5	1.2
9	29	Female	0.8	0.2	160	35	25	5.8	2.9	0.9
10	40	Male	2.3	1.0	220	60	70	6.9	3.8	1.3
11	50	Female	1.2	0.4	190	30	40	7.2	4.0	1.5
12	27	Male	0.7	0.1	175	50	45	6.3	3.1	1.0
13	42	Male	0.7	0.2	197	64	33	5.8	2.4	0.7
14	69	Male	0.9	0.2	215	32	24	6.9	3.0	0.7
15	53	Male	1.6	0.9	178	44	59	6.5	3.9	1.5
16	49	Male	1.3	0.4	206	30	25	6.0	3.1	1.06

B.5 Audit Trace: Fold 5 | Ctx: 20% | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	60	Male	11.0	4.9	750	140	350	5.5	2.1	0.6
2	37	Male	1.8	0.8	145	62	58	5.7	2.9	1.0
3	38	Male	0.8	0.2	208	25	50	7.1	3.7	1.0
4	45	Female	0.6	0.1	270	23	42	5.1	2.0	0.5
5	13	Male	1.5	0.5	575	29	24	7.9	3.9	0.9
6	18	Male	0.6	0.2	538	33	34	7.5	3.2	0.7
7	29	Female	0.9	0.3	210	35	29	6.5	3.2	1.1
8	34	Male	1.5	0.6	300	45	50	7.2	3.8	1.3
9	47	Female	2.1	0.9	400	60	75	6.8	3.5	1.0
10	52	Male	0.8	0.2	150	25	30	7.0	3.0	1.2
11	39	Female	1.2	0.4	280	55	65	6.9	3.4	0.9
12	44	Male	2.5	1.1	320	70	80	7.4	3.6	1.1
13	33	Male	2.1	0.7	205	50	38	6.8	3.0	0.7
14	22	Female	6.7	3.2	850	154	248	6.2	2.8	0.8
15	48	Male	5.8	2.5	802	133	88	6.0	2.8	0.8
16	84	Female	0.7	0.2	188	13	21	6.0	3.2	1.1

B.6 Audit Trace: Fold 6 | Ctx: 20% | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	60	Male	3.2	1.8	750	79	145	7.8	3.2	0.69
2	45	Male	2.2	1.6	320	37	48	6.8	3.4	1.0
3	39	Male	1.9	0.9	180	42	62	7.4	4.3	1.38
4	60	Male	22.8	12.6	962	53	41	6.9	3.3	0.9
5	70	Male	3.1	1.6	198	40	28	5.6	2.0	0.5
6	42	Male	0.8	0.2	127	29	30	4.9	2.7	1.2
7	34	Female	1.2	0.4	250	45	60	6.5	3.1	0.85
8	29	Male	0.9	0.3	180	30	40	7.0	4.0	1.0
9	52	Female	2.5	1.0	300	50	70	6.8	3.5	0.75
10	41	Male	3.0	1.5	400	80	90	7.2	3.8	0.95
11	38	Female	1.1	0.5	220	35	50	6.3	3.2	0.78
12	45	Male	4.0	2.0	500	100	110	7.5	4.1	0.85
13	50	Male	0.9	0.3	194	190	73	7.5	3.9	1.0
14	49	Male	0.6	0.1	218	50	53	5.0	2.4	0.9
15	18	Male	0.8	0.2	228	55	54	6.9	4.0	1.3
16	30	Female	0.7	0.2	194	32	36	7.5	3.6	0.92

B.7 Audit Trace: Fold 7 | Ctx: 20% | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	42	Male	8.9	4.5	272	31	61	5.8	2.0	0.5
2	57	Male	4.0	1.9	190	45	111	5.2	1.5	0.4
3	21	Female	0.6	0.1	186	25	22	6.8	3.4	1.0
4	32	Male	0.7	0.2	165	31	29	6.1	3.0	0.96
5	40	Male	3.5	1.6	298	68	200	7.1	3.4	0.9
6	37	Male	0.7	0.2	235	96	54	9.5	4.9	1.0
7	29	Female	1.5	0.5	150	22	30	6.0	3.0	1.0
8	45	Male	3.2	1.2	200	40	50	7.0	3.5	0.9
9	38	Female	0.8	0.3	180	25	35	6.5	3.2	1.1
10	52	Male	2.5	1.0	250	30	40	7.5	3.8	0.8
11	34	Female	1.0	0.4	170	20	28	6.2	3.1	1.0
12	60	Male	4.0	1.5	300	45	60	8.0	4.0	0.75
13	13	Female	0.7	0.2	350	17	24	7.4	4.0	1.1
14	54	Female	23.2	12.6	574	43	47	7.2	3.5	0.9
15	63	Male	0.9	0.2	194	52	45	6.0	3.9	1.85
16	46	Male	0.6	0.2	290	26	21	6.0	3.0	1.0

B.8 Audit Trace: Fold 8 | Ctx: 20% | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	49	Male	0.6	0.1	218	50	53	5.0	2.4	0.9
2	51	Male	2.9	1.2	189	80	125	6.2	3.1	1.0
3	60	Male	2.6	1.2	171	42	37	5.4	2.7	1.0
4	65	Male	0.7	0.2	265	30	28	5.2	1.8	0.52
5	4	Male	0.8	0.2	460	152	231	6.5	3.2	0.9
6	18	Female	0.8	0.2	199	34	31	6.5	3.5	1.16
7	45	Male	1.2	0.4	210	55	60	6.0	3.0	1.0
8	29	Female	0.5	0.1	180	25	30	6.5	3.2	1.1
9	37	Male	2.1	0.9	300	70	80	7.0	3.5	1.2
10	52	Female	0.8	0.2	220	40	35	6.8	3.4	1.0
11	60	Male	3.0	1.5	250	90	75	7.5	3.8	1.1
12	34	Female	1.5	0.7	190	30	28	6.2	2.9	1.0
13	48	Male	0.7	0.2	208	15	30	4.6	2.1	0.8
14	60	Male	22.8	12.6	962	53	41	6.9	3.3	0.9
15	27	Male	1.0	0.3	180	56	111	6.8	3.9	1.85
16	28	Female	0.8	0.2	309	55	23	6.8	4.1	1.51

B.9 Audit Trace: Fold 9 | Ctx: 20% | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	60	Male	5.8	2.7	599	43	66	5.4	1.8	0.5
2	55	Male	1.1	0.3	215	21	15	6.2	2.9	0.8
3	33	Male	2.0	1.0	258	194	152	5.4	3.0	1.25
4	28	Female	0.9	0.2	316	25	23	8.5	5.5	1.8
5	33	Male	2.1	0.7	205	50	38	6.8	3.0	0.7
6	35	Female	0.9	0.2	190	40	35	7.3	4.7	1.8
7	34	Female	1.2	0.5	210	35	30	6.0	3.5	1.0
8	29	Male	0.8	0.3	180	40	20	7.0	4.0	1.2
9	45	Female	2.5	1.1	250	50	45	7.5	3.8	0.76
10	52	Male	1.5	0.7	300	60	50	6.8	3.2	0.53
11	38	Female	0.9	0.4	220	30	25	6.5	3.0	0.75
12	41	Male	3.0	1.5	400	70	60	8.0	4.5	0.56
13	26	Male	2.0	0.9	195	24	65	7.8	4.3	1.2
14	47	Male	2.7	1.3	275	123	73	6.2	3.3	1.1
15	74	Male	1.0	0.3	175	30	32	6.4	3.4	1.1
16	48	Male	3.2	1.6	257	33	116	5.7	2.2	0.62

B.10 Audit Trace: Fold 10 | Ctx: 20% | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	49	Male	3.9	2.1	189	65	181	6.9	3.0	0.7
2	60	Male	0.7	0.2	171	31	26	7.0	3.5	1.0
3	27	Male	0.6	0.2	161	27	28	3.7	1.6	0.76
4	39	Male	1.9	0.9	180	42	62	7.4	4.3	1.38
5	48	Female	0.9	0.2	175	24	54	5.5	2.7	0.9
6	57	Male	1.4	0.7	470	62	88	5.6	2.5	0.8
7	34	Female	1.2	0.4	210	30	40	6.5	3.2	0.85
8	29	Male	0.7	0.2	150	25	30	5.9	2.9	0.75
9	55	Female	3.5	1.5	300	45	60	7.2	3.8	0.9
10	42	Male	0.9	0.3	180	35	50	6.0	3.0	0.8
11	38	Male	2.1	0.9	220	40	55	6.8	3.4	0.95
12	60	Female	1.0	0.5	190	32	45	6.3	3.1	0.78
13	54	Female	5.5	3.2	350	67	42	7.0	3.2	0.8
14	43	Male	22.5	11.8	143	22	143	6.6	2.1	0.46
15	57	Male	4.5	2.3	315	120	105	7.0	4.0	1.3
16	65	Male	0.7	0.2	199	19	22	6.3	3.6	1.3